FabriCAD

Status report — Thursday 22 October 2026, 10:00

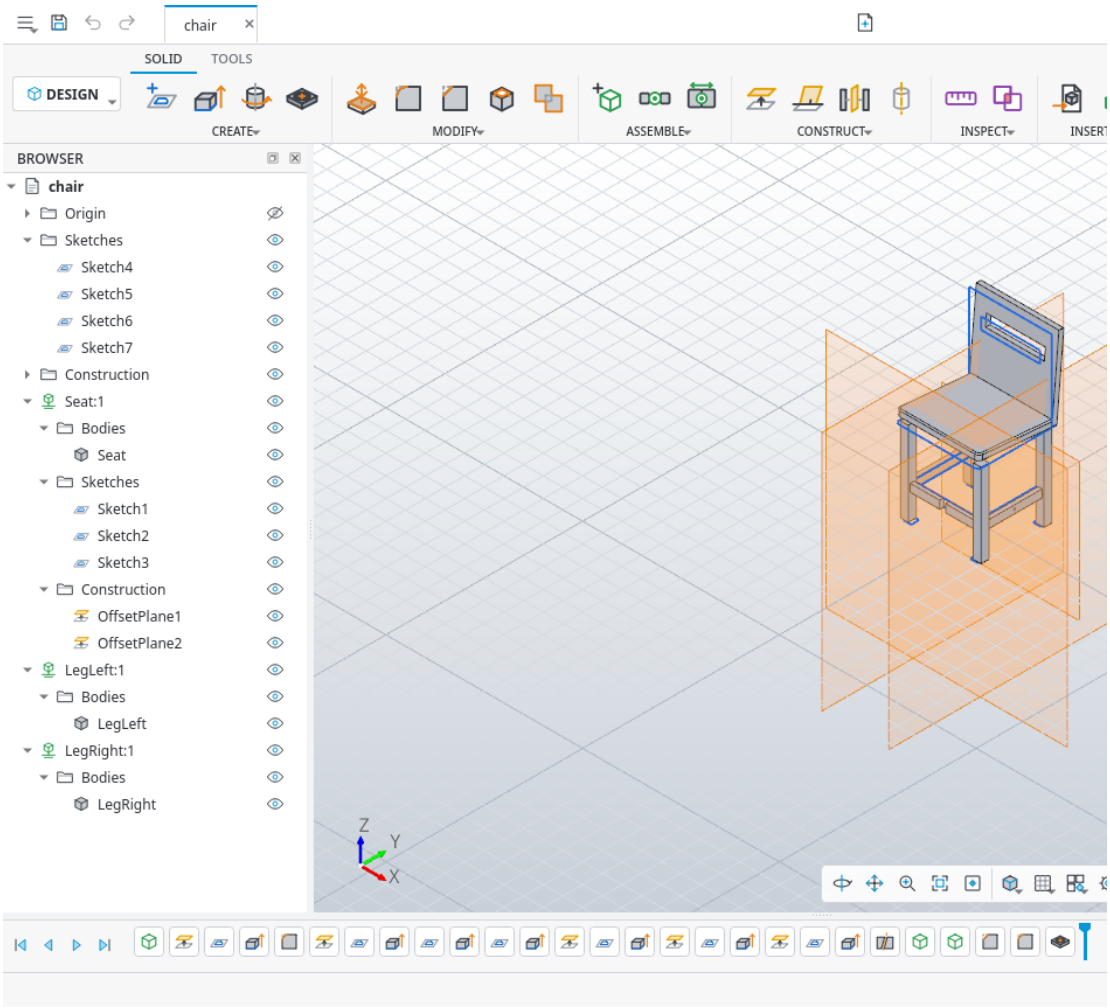
FabriCAD is an open-source parametric CAD application (C++20, Qt 6, OpenCASCADE 7.9) with a Fusion 360–style interface for parts, assemblies and engineering drawings. This report covers the state of the app after the overnight test-and-fix sequence: what was built and fixed, what has been tested, what is still untested, and what is missing from the current scope.

57/57 automated tests passing (Release build)	**129** defects triaged in 3 overnight test rounds	**119** verified fixed on retest (+22 partial)	**0** regressions found on retest
~108k lines of C++ source	**127** commits on main (nothing pushed)	**0.22 s** window startup (FreeCAD 1.1.4: 2.64 s)	**2** issues retested as not fixed

At a glance

- **Usable end to end:** sketch > features > assembly with joints > drawing > STEP/STL/PDF/SVG/DXF export. The 3-part chair benchmark builds exactly to spec (volumes within 0.001 mm³) and is a regression test; an 84 s recording of it is at *docs/uiux/chair-run.mp4*.
- **Overnight:** UI/UX, functional/reliability and general-bugs rounds each ran test > triage > plan > fix > merge > retest, with Sonnet testers driving the real GUI using only mouse and keyboard.
- **Not reached:** the dedicated drawing round could not start before the 09:00 cutoff, so the Drawing workspace is the least-tested area.
- **Your reported issues:** light-on-light text, eye toggles needing a redraw and unselectable faces/edges are fixed. The ViewCube now drags to orbit with larger hit zones, but is still low-contrast and corner clicks are unreliable.

1. Current app state



The chair benchmark open in the current build: browser with components, bodies, sketches and construction planes; timeline with every feature; Fusion-style toolbar.

Area	State
Build	Release build in *build/*, 0 warnings; 57/57 ctest entries pass (unit, GUI-logic, end-to-end, chair benchmark, scripting). Run with *build/bin/fabricad*.
Interface	Light theme across menus, dialogs, popups; tabbed ribbon (SOLID/SKETCH/ASSEMBLE/CONSTRUCT/INSPECT/INSERT); browser, timeline, ViewCube, nav bar, S search, cmd> line, marking menu, document tabs; 233 SVG icons.
Modelling	Constraint-driven sketcher (PlaneGCS) with on-screen value boxes; parametric history with rollback, edit, suppress; drag manipulators on features.
Assemblies	Components, ground, 6 joint types with drive, as-built joints, move/rotate triad, copy/paste instances, Paste New, component patterns, interference, BOM.
Drawings	Sheets and title block, base/projected/section/detail/break views, dimensions, tolerances, GD&T, callouts, parts list and balloons; PDF/SVG/DXF export.
Automation	Headless *--script* mode (same command line as the GUI) with *--step-delay* for watchable runs; UI test harness *tools/uitest/ui.sh*.
Performance	Window in ~0.22 s (FreeCAD 2.64 s); chair script 2.6 s headless (FreeCAD 1.3 s, mainly because every typed value recomputes the live preview).

2. Features added and fixed

Stage	What was added or fixed
Round 1 (build)	Core document/timeline/undo, sketcher + solver, part features (extrude, revolve, sweep, loft, fillet, chamfer, shell, hole, patterns, mirror, combine, press-pull, planes/axes/points), assemblies with joints, drawings, Fusion-style shell, STEP/STL I/O.
Round 2 (features)	Icons; sketch value boxes that create dimensions; on-canvas manipulators; document tabs and scripting mode; Split Body, primitives, threads, tapped holes, variable fillet, rib, thicken, scale, Paste New, component patterns; ellipses, splines, sketch text, sketch patterns; drawing tolerances, GD&T, callouts, break views.
Chair benchmark	7 defects fixed: scoped auto-join, fresh STL meshing, holes cut only their own body, safe component delete, IMPORT path, STEP part names, Export STL dialog; pivot rotation.
UI/UX round	41 issues: full light theme and readable text; viewport redraws immediately (eye toggles); picking of faces/edges/bodies restored; ViewCube drag-to-orbit, larger hit zones, hover, home/roll/step arrows; dialogs dock top-left; fit margins; labelled origin planes.
Functional round	39 issues: per-instance autosave recovery, clear load/feature-failure messages, delete asks about dependents, pattern count capped, over-thick shell handled, assembly and sketch reliability fixes.
General round	49 issues: closing a design closes its drawings, sheets panel stays in sync with undo, failed save keeps name, suppressed inputs mark dependents as errors, modified flag follows undo, thread cutter, expression errors block OK, ellipse/spline/text tool fixes.

Retest results by area

Counted from the retest tables in *docs/uiux/retest* and *docs/tests/*/retest*.

Round	Area	Fixed	Partial	Not fixed	New (minor)
UI/UX	Viewer	**13**	4	1	4
UI/UX	Sketch	**15**	2	0	5
UI/UX	Solid	**13**	0	0	7
UI/UX	Assembly + drawing	**5**	4	0	6
Functional	Sketch	**10**	1	0	0
Functional	Part	**7**	0	0	3
Functional	Documents	**7**	1	0	2
Functional	Assembly	**9**	3	0	2
General	Advanced part	**6**	1	0	4
General	Advanced sketch	**13**	2	1	3
General	App tools	**10**	1	0	3
General	Bug hunt	**11**	3	0	2
Total		**119**	**22**	**2**	**41**

3. Features tested

All overnight testing was done through the real GUI with mouse and keyboard only, on private X displays, and observed through screenshots. Earlier rounds also added automated tests and a scripted chair benchmark.

Area	Tested and working	Result
Sketch	Lines, rectangles, circles, arcs, polygons, slots, ellipses, splines, text, patterns, move/rotate/scale, trim/extend/break/offset/fillet/chamfer, project, constraints, dimensions, edge cases (zero-size, 0.01 mm-10 m, conflicts), sketch on angled faces	Good
Part features	Extrude (all modes/extents), revolve, fillet (constant/variable), chamfer, shell, hole (simple/counterbore/tapped), patterns, mirror, combine, press-pull, primitives, threads, draft, sweep, loft, rib, thicken, scale, split, planes/axes/points	Good
History	Timeline edit, rollback, insert, suppress, delete with dependents, long undo/redo	Good
Documents	Save/open fidelity (34 features), Save As, tabs, autosave recovery after kill -9, corrupt files, STEP/STL export and import	Good
Assemblies	Components, ground, rigid/revolute/slider joints, drive, move/rotate triad, copy/paste, nested components, split-then-components	Mostly good
App tools	Measure, interference, materials, appearance, parameters with units, S search, cmd> line, marking menu, shortcuts, help, context menus	Good
Viewer / UI	Navigation, ViewCube, display modes, selection and filters, visibility, theme and contrast	Mostly good
Benchmark	3-part chair vs FreeCAD 1.1.4: exact geometry, watertight STL, STEP part names, timing	Passes
Drawings	Only touched incidentally (base view, sheet add/delete/undo, PDF export in early tests)	Thin

Still open after retest

- ViewCube is still faint on a light background, and a click near a corner sometimes gives an edge view (UX-05).
- Origin child items have no eye toggles (UX-40). Origin planes can be drawn oversized and look doubled (UX-10).
- After Finish Sketch no profile shading is shown (SK-12, not fixed).
- Selection: no window/freeform/paint modes, and no check mark on the active filter (UX-39).
- Rectangle value boxes can sit under the cursor and swallow the second-corner click (new, medium).
- Joint dialog layout and the Dimension hint were not retested. The extrude value box does not follow the handle after an orbit.

4. Features not yet tested

- **Drawing workspace in depth:** projected, section, detail and break views; all annotation types; associativity after model edits; parts list and balloons; quality of the exported PDF/SVG/DXF files. The dedicated drawing round is prepared and ready to run.
- **HiDPI and scaling:** nothing was tested at 2× device pixel ratio, and the ViewCube size is in device pixels.
- **Native Wayland:** the system OpenCASCADE is X11/GLX only, so all testing ran through XWayland.
- **Scale:** large assemblies (hundreds of parts) and very long histories, beyond the 34-feature model.
- **Round-trips with other CAD:** STEP files from other CAD systems, and files from FreeCAD or Fusion.
- **Live GUI timing:** click and keystroke counts versus FreeCAD were estimated from code, not measured.

5. Missing features (within the current scope)

The scope is a fast, lightweight, Fusion-like parametric modeller for parts, assemblies and drawings. These gaps were found by testers, in reports, or by checking the source.

Area	Missing or incomplete
Application	Preferences dialog (testers found none); dark theme; HiDPI tuning; separate headless runner without the main window.
Sketch	Sketch-level profile shading after Finish Sketch; offset dimension for curves; constraint for point-to-curve distance; absolute coordinates while drawing create no constraints.
Part	Sheet metal; surface-modelling workspace; boundary fill; section analysis and zebra inspection; manipulators for sweep, loft, draft, patterns, thread and split.
Assembly	Motion links, contact sets and motion study; exploded views; assembly structure kept on STEP import (it is flattened into bodies).
Drawing	Exploded and animated views; DXF import into sketches; depth of standards compliance not verified.
Selection / UI	Window/freeform/paint selection modes; eye toggles on Origin children; ViewCube contrast.
Performance	Scripted builds are about 2× slower than FreeCAD (preview recomputes per typed value, full recompute on delete); the default STL tessellation is coarser than FreeCAD's.

6. Where things are

Item	Location (in ~/Documents/projects/fabricad)
App binary	build/bin/fabricad (--script, --step-delay, --offscreen)
Test reports	docs/uiux/, docs/tests/functional/, docs/tests/general/ (round1, plans, fixes, retest)
Benchmark	docs/comparison/COMPARISON.md, CHAIR_SPEC.md, tests/chair/
Recording	docs/uiux/chair-run.mp4 (84 s)
Next step	Run the drawing round (args ready in the session scratchpad test-plan.json)